State-Driven Retrieval and Learned Re-Ranking for Conversational Music Recommendation

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Abstract

We describe team npatta01's submission to the RecSys Challenge 2026 conversational music recommendation task: given a multi-turn dialog, retrieve 20 tracks from a 47,071-track catalog and generate a natural-language response. The pipeline extracts a typed conversation state with an LLM, gathers candidates from eleven retrieval branches over a unified track index, re-ranks the pooled candidates with a LambdaMART model whose dominant feature is a fine-tuned conversational bi-encoder cosine, and generates a state-conditioned response. On the final Blind-B leaderboard the submission scored 0.3811 composite (nDCG@20 0.2537, LLM-judge 3.30), ranking 29th of 39 teams. We report the system in full, and then examine our own result: the in-sample development estimates that guided development overstated performance, while leakage-safe out-of-fold estimates (nDCG@20 0.197–0.203) anticipated the blind outcome; the benchmark's single-ground-truth labels carry a quantified *anchoring bias* that rewards copying the just-played artist; and failure cases from the submitted run show extracted constraints the pipeline could not enforce. Code, models, and a full reproduction bundle are publicly released.

CCS Concepts

• **Information systems → Recommender systems**; *Retrieval models and ranking*; *Music retrieval.*

Keywords

conversational recommender systems, music recommendation, learning to rank, reciprocal rank fusion, label noise, RecSys Challenge

1 Introduction

The Music-CRS challenge [1, 2] asks systems to act as a conversational music recommender. Each session is a multi-turn dialog between a listener and an assistant; at every assistant turn the system must (i) retrieve a ranked list of 20 tracks from a shared 47,071-track catalog and (ii) write a short natural-language response presenting its top recommendation. Training data comprises 15,199 sessions built on the TalkPlayData 2 collection [3], with per-track metadata (title, artist, album, up to 37 genre/mood tags, popularity, release date) and precomputed embeddings (text, audio, cover art, collaborative filtering). Evaluation is on two 80-session blind splits of final turns; Blind-B additionally removes the conversation-goal annotations present in training data. Submissions are scored with the organizers' composite metric:

$$0.50 \cdot \text{nDCG@20} + 0.10 \cdot \text{cat. div.} + 0.10 \cdot \text{lex. div.} + 0.30 \cdot \frac{\text{judge}-1}{4},$$

where "judge" is a 1–5 rating of the generated response from the organizers' LLM-based evaluator. In addition to the training sessions, a 1,000-session development split with public per-turn ground truth (one target track per turn) supports offline iteration.

Our system is a retrieve-then-rerank pipeline conditioned on an explicitly compiled conversation state. This paper documents the pipeline (Section 2) and its results (Section 3), then analyzes the errors behind the final score (Section 4): an evaluation mistake in which in-sample development replays overstated performance while leakage-safe out-of-fold estimates anticipated the Blind-B result; a fine-tuned conversational bi-encoder whose single cosine feature carries 59% of the reranker's decision weight and displaces a label-noise-induced artist-copy shortcut; and a quantified anchoring bias in the ground-truth labels, illustrated with failure cases from the submitted run. Code, models, and a cleaned, LLM-re-judged label set are released at https://github.com/npatta01/music-crs-2026.

Related work in brief: the design follows the retrieve-then-rerank pattern standard in large-catalog recommendation, with reciprocal-rank fusion [4] for multi-source pooling, LambdaMART [5, 6] for learned ranking, and LLM-based dialog state tracking.

2 Approach

2.1 State extraction

At each turn we prompt `deepseek-v4-flash` with the full conversation so far (all prior turns plus the user profile) to emit a schema-constrained conversation state. Its eleven fields cover the current request and intent mode, track feedback and pinned references, mentioned entities, hard filters and explicit rejections, process constraints, routing flags, lyrical theme, and a release-year window; Figure 2 shows the schema populated on a real turn. A fuzzy resolver grounds surface names to catalog IDs so downstream stages operate on identifiers. Cheaper extraction models under-filled optional fields (rejections, attribute facets) in our checks, so all final runs use `deepseek-v4-flash`. Extraction runs once per turn and is cached.

2.2 Candidate retrieval

All per-track information lives in a **single LanceDB collection**: metadata fields, a full-text index, and every embedding column — three Qwen3-Embedding text views (metadata, attributes, lyrics; 1024-d), LAION-CLAP audio (512-d) [8], SigLIP-2 cover art (768-d) [9], BPR collaborative-filtering vectors (128-d) [10], and our bi-encoder document vectors (2560-d, Section 2.4). Every retrieval branch is a differently-shaped query against the same table — in principle the whole retrieval step could be one composed query clause (Section 4.3).

Table 1 lists the eleven branches that produce candidate pools each turn. In the BM25 branch [11], free-text mood and genre phrases first pass through a **tiered tag resolver**: exact, alias, and

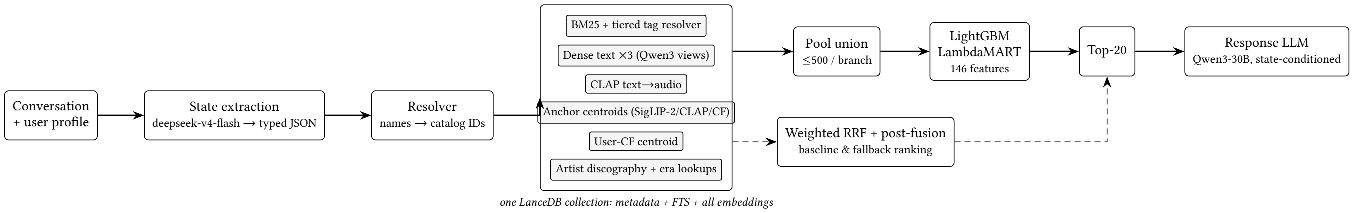

Figure 1: System overview. The conversation is compiled into a typed state; eleven retrieval branches query one LanceDB collection; the LambdaMART reranker scores the branch-pool union and emits the top-20; a response LLM presents the top track. Weighted RRF over the same branches provides the baseline and fallback ranking (dashed).

User (turn 3): "…subdued, female singer, a sense of place in the lyrics, stark almost a cappella delivery…not it either."

Extracted state (abridged):

```
request_type: hidden_target
anchor artist: Neko Case (must_use)
rejected tracks: "Calling Cards", "Man"
facets: mood=subdued; lyrical_theme=sense of place;
        sonic=stark almost a cappella
```

Top-1: Neko Case — "Bracing For Sunday"

Submitted response: "You're looking for something subdued with a strong sense of place and a stark, almost a cappella delivery—and *Bracing For Sunday* fits that mood perfectly. . . . "

Figure 2: A real Blind-B turn (session `024a2738...`): extracted state and submitted output for a hidden-target request.

Table 1: The eleven retrieval branches (all queries against the one collection).

Branch	Space	Query
BM25 (fielded)	lexical	state text + resolved tags
Dense text ×3	Qwen3 views	state-built strings
CLAP text→audio	audio	sonic description
Anchor centroids ×3	SigLIP-2/CLAP/CF	liked-track centroids†
User-CF centroid	CF-BPR	the user's own vector
Lookups ×2	catalog	discography; era popularity

† Gated by intent mode: a pivot suppresses anchor centroids.

substring matching against the catalog tag vocabulary, with an embedding nearest-neighbor fallback (cosine ≥ 0.6, top-3). Weighted reciprocal-rank fusion [4] (k=60, top-1000) plus post-fusion multiplicative adjustments (rejected artists/tracks dropped, played tracks dropped, exact-membership tag promotes/demotes, year-range decay) yields an explicit ranking that serves as **baseline and fallback**.

2.3 Learned re-ranking

The submitted system replaces the fused order entirely: a LightGBM LambdaMART [5, 6] model scores the **union of the branch pools** (each truncated to its top 500) and emits the final top-20. It sees 146 features per (turn, track) pair: per-branch rank/score/margin/hit features, pool-normalized z-scores and percentiles, content and behavioral cosines, session/artist-affinity counters, tag and lexical overlap, popularity, extracted-state categoricals, and a constraint sidecar. Table 2 groups them by family with each family's share of total split gain in the deployed model; one feature — the bi-encoder cosine `b1_cos` — carries 59.0% of all gain. The RRF rank itself is *excluded*: the ranker sees raw branch evidence, not the fused opinion. After scoring, two rule-based guards apply: an exact-track pin when the user names a specific track, and a final-artist constraint check (enabled for the Blind-B run).

Table 2: Reranker feature families (share of total gain, deployed model).

Family	#	Gain	Examples
Bi-encoder cosine	1	59.0%	`b1_cos`
Per-branch rank/score	66	11.5%	BM25 rank, dense margins
Session/artist affinity	18	10.7%	same-artist, played count
Other similarity cosines	17	8.9%	CLAP/SigLIP centroids
Popularity	7	5.0%	pop. percentile
State/intent	14	2.2%	request type, intent mode
Tag/lexical overlap	10	2.0%	IDF tag overlap
Constraints/rejections	13	0.7%	rejected-artist flag

Training uses the per-turn ground-truth track as a binary label (LambdaRank objective optimizing nDCG@20; learning rate 0.025; 127 leaves; truncation 200; 5-fold user-grouped cross-validation; ∼20.6M rows). Because the ground truth is noisy (Section 4.2), labels are down-weighted when the *next* turn contradicts them: ×0.3 if goal progress says the recommendation did not move toward the goal, ×0.3 if the user rejects the ground-truth track, ×0.6 if they reject its artist. The deployed bundle is *goal-free*: trained without the conversation-goal fields Blind-B removes. Cost shaped the training set: state extraction and per-turn retrieval tracing run through a paid LLM API, so the model trains on a 30,000-turn subset (∼25% of the ∼121.6k-turn split); the data-scaling curve had not flattened at that size.

2.4 The bi-encoder feature `b1_cos`

The reranker's dominant feature comes from a two-tower bi-encoder fine-tuned from Qwen3-Embedding-4B [7]. The query tower renders the conversation compactly behind a retrieval instruction prefix and the document tower renders a track card (both formats in Figure 3); the card's one-line LLM-written "known for" artist description imports static world knowledge from larger models into the embedding space. One shared encoder E_θ serves both towers; with $v_x = E_\theta(x)/\|E_\theta(x)\|_2$ (last-token pooling, 2560-d), the feature is the cosine $\texttt{b1\_cos}(q,d) = v_q^\top v_d$. Training minimizes the in-batch contrastive (MNRL [13]) loss over batches of positive pairs (q_i, d_i^+)

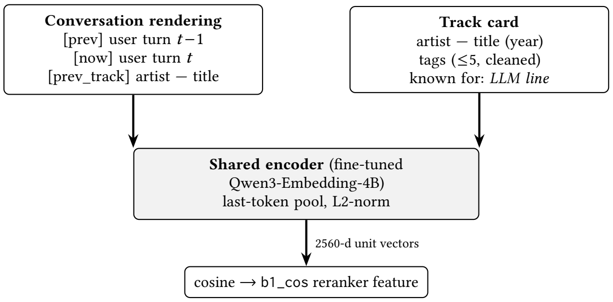

Figure 3: The b1 bi-encoder: one shared fine-tuned encoder embeds the conversation and the track card; their cosine is the reranker's dominant feature.

with four mined hard negatives per positive,

$$\mathcal{L} = -\frac{1}{B}\sum_i \log \frac{\exp(v_{q_i}^\top v_{d_i^+}/\tau)}{\sum_{d\in\mathcal{D}_i}\exp(v_{q_i}^\top v_d/\tau)}, \qquad \tau = \frac{1}{20},$$

where $\mathcal{D}_i$ contains the in-batch documents plus the hard negatives. Positives are conservative: the 53,885 (turn, track) pairs whose next-turn goal-progress annotation says the recommendation *moved the session forward*. Document vectors for all 47,071 tracks are precomputed into the collection; the conversation vector is encoded live with caching.

We deploy the bi-encoder as a reranker feature only: as a retrieval branch it added little coverage, but its cosine became the top signal (Table 2) — and a *generalizing replacement* for a shortcut learned from biased labels (Section 4.2): in leakage-safe out-of-fold ablations, removing the artist-consensus shortcut features reduces nDCG@20 by 0.0070, while adding `b1_cos` more than recovers the loss (+0.0091) with the shortcut absent.

2.5 Response generation

The response generator is `Qwen3-30B-A3B-Instruct-2507` [12] at temperature 0: a single pass over the top-1 track rendered as a delimited XML block (≤10 tags; the delimiting stops the model from echoing raw metadata) plus the *latest* extracted state. The deployed template's full style instruction is:

> *"Write 1–2 concise sentences about only the selected track. Prioritize the latest user request and extracted state over older conversation history. If the track is reasonably aligned, explain the fit with one specific supported reason. If it clearly conflicts with an explicit avoid/new-artist constraint, do not oversell it or blame the system; briefly frame the limitation and the closest supported reason."*

Template choice matters: over a frozen Blind-A retrieval output, a sweep of eight templates — varying state conditioning, history depth, item rendering, and concision — moved the judge score from 3.95 to 4.70 without touching the recommendations. There is no multi-draft sampling, selection, fact-checking, or repair pass.

Code, configs, trained models, cached extraction states, and a frozen-LLM-cache reproduction bundle that replays our submissions without API credentials are public.[1]

[1]https://github.com/npatta01/music-crs-2026; models and data: https://huggingface.co/datasets/Npatta01/music-crs-repro-2026.

Table 3: Development split, with evaluation lineage. The in-sample row replays turns the deployed (full-data) model saw during training; the out-of-fold rows are leakage-safe cross-validation estimates.

System (evaluation lineage)	nDCG@20
Weighted RRF fusion (no training)	0.1492
LambdaMART, out-of-fold CV	0.1970
+ `b1_cos` (deployed feature set)	0.2032
LambdaMART, evaluated in-sample (train replay)	0.3844

Official Blind-B nDCG@20: 0.2537 — between the out-of-fold estimates and far below the in-sample replay.

Table 4: Official CodaBench results; the composite formula is defined in Section 1. The Blind-A rank counts all development-phase submissions.

Split	nDCG	Cat.	Lex.	Judge	Comp.	Rank
Blind-A (dev)	0.4380	0.0313	0.7670	4.20	0.5389	63/181
Blind-B (final)	0.2537	0.0315	0.7862	3.30	**0.3811**	**29/39**

3 Results

The development split (1,000 sessions × 8 turns) has one ground-truth track per turn, so Recall@k equals Hit@k. Table 3 reports development-split numbers *with their evaluation lineage made explicit*; Table 4 reports the official leaderboard results.

Learned re-ranking beats static fusion by the leakage-safe estimate (0.149 → 0.197–0.203); the in-sample replay (0.384) overstated it, and the blind result (0.2537) tracked the out-of-fold evidence available before submission.

4 Error Analysis

4.1 In-sample development evaluation

Because every training and evaluation turn needs LLM state extraction and full retrieval tracing, building large held-out evaluation sets was expensive, so the development split served two roles: its turns entered training, and evaluation replayed the same turns. The deployed reranker was fit on all of its feature data (CV folds informed early stopping, then a full-data model shipped), so the headline 0.38–0.46 development estimates were in-sample, and they drove selection confidence. The leakage-safe out-of-fold numbers (0.197–0.203), which we had, predicted the blind outcome (0.2537) far better. We regard this as a measurement error rather than a demonstrated cause of the final ranking, although it influenced which directions appeared sufficiently solved. The practical lesson is to base model selection on out-of-fold estimates.

4.2 Ground-truth label noise

We encountered this problem while attempting to build higher-quality training data: for many turns, the labeled target contradicts what the user had just asked. The benchmark's per-turn ground truth is a *sibling pick* from the listener's real session — the track actually played next — not an editorial judgment of the request. In a manual sample only 62% of ground-truth tracks were a clear

User: "…something with a similar chill, electronic vibe, but **from a different artist**?" (just played: *Bonobo*)
Ground truth: Bonobo — "Jets" *annotation: MOVES toward goal*
User (next turn): " 'Jets' is cool, but I was actually hoping for something **from a different artist this time**…"
Ground truth: Bonobo — "Pieces" *annotation: MOVES toward goal*

Figure 4: Anchoring bias in the training labels (real session): two consecutive explicit different-artist requests, yet the ground truth stays on Bonobo and the synthetic reaction claims the listener liked it.

Table 5: Label-free audit of the submitted Blind-B run: where the flagged turns were lost.

Diagnosis	Turns
Better candidate in pool, ordered below top-20	31
Over-aggressive filtering	8
State-extraction miss	3
Entity-resolution miss	1

fit to the request, 26% a loose fit, and 11% no fit. The dominant pattern is what we call **anchoring**: the ground truth stays with the just-played artist even against an explicit request for someone else (Figure 4).

To measure it we re-judged all 106,393 training turns (and 7,000 development turns) with an LLM pipeline: two inexpensive judges score each (request, ground-truth) pair on two axes — did the user ask for a different artist, and does the track fit the request — and a stronger arbiter re-judges conflicts *blind to the synthetic reaction annotation*. Results: 57.4% of turns judged negative; 18,222 are anchoring negatives; 5,880 of those carry a synthetic "listener liked it" annotation — poisoned positives for any goal-progress-based training scheme, including our bi-encoder's. A second pattern — the user rejects the ground-truth track on the next turn — motivated our ×0.3 label down-weighting. Anchored labels teach a specific wrong lesson: *copy the last artist*. Our ablations show the ranker exploiting exactly this, and the bi-encoder cosine replacing the shortcut (Section 2.4). Since every team is scored against the same labels, the bias depresses and partially reshuffles all measured scores. We release the cleaned relabeling with our code; the submitted models were *not* trained on it.

4.3 Failure cases

A separate label-free audit — a single LLM judge over the 80 submitted Blind-B turns, unlike the two-judge pipeline of Section 4.2 — rated 68% of them weak-or-bad fits; Table 5 breaks the flagged turns down by pipeline stage (hidden-label frequencies remain unknown). Concrete cases from the submitted predictions, each with the systemic gap it exposes:

- **Anchoring at serving time.** "Another alt-country song… but **from a different artist than Ryan Adams**" → our top-1 was *Ryan Adams & The Cardinals*, plus five more of their tracks in the top-20. The rejection resolved to the solo "Ryan Adams" catalog ID; the band-variant ID passed the hard-drop. *Gap: rejections are enforced by exact catalog ID, but artist identity is fragmented across band variants and duplicate entries — a name-level veto would have caught this. The system also reproduced exactly the bias of Section 4.2.*
- **Rejected artist in the list.** "Symphonic black metal with strong **female operatic vocals**, **not Cradle of Filth**" → four Cradle Of Filth tracks in the top-20 (a duplicate artist-ID variant escaped the drop), and the top-1 was male-fronted power metal. *Gap: the same identity fragmentation, plus no vocal-attribute field to enforce.*
- **Impossible exact request.** "Play 'Watercolors' by Pat Metheny" — the track does not exist in the 47k catalog; we returned Pat Metheny's "Alfie" without saying so. *Gap: the system carries no world knowledge or external data with which to recognize an out-of-catalog request, and the single-pass response cannot flag unavailability; the response term carries 30% of the composite, where our submission lost the most points (judge 3.30).*
- **Unactionable constraint.** "Any aggressive metal track that **exactly matches 126.70 bpm and G minor**, with tempo and key stated" — the catalog has no BPM or key fields. *Gap: the same constraint gap in a harder form — the state captured the constraint, but the catalog offers no field to map it onto.*

Two further gaps sit behind these cases. First, *constraints were captured but not mapped to filters*: the state records what the user asked for, yet few of those constraints became executable predicates — each branch consumed only a slice of the state — so candidate recall stayed noisy; tag handling illustrates it (the BM25 clause resolves free-text moods to canonical tags, yet post-fusion promotes/demotes and the reranker's overlap features use exact string membership, and only `tag_emb_cos` sees semantic similarity). Second, *missing evidence lanes*: the pipeline had no track-to-track co-occurrence or sequential-transition source (behavioral signal entered only through CF centroids and lookups) and no live reasoning over candidates (LLM world knowledge entered only through the offline "known for" lines, though hidden-target requests like Figure 2 demand it) — and no LightGBM column can recreate a signal that never entered the pipeline.

5 Conclusion

We documented a state-compiled retrieve-then-rerank conversational music recommender in full and audited our mid-pack result: an in-sample development estimate inflated confidence that leakage-safe evidence had already contradicted; the ground truth carries a quantified anchoring bias our own submission reproduced at serving time; and several extracted constraints had nothing in the catalog to act on. We release the pipeline, the models, a replay bundle, and a cleaned relabeling of the training turns.

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